

DAMA Chapter 4: Data Architecture

Presentation overview with concepts, examples, and visual summaries

Prepared for Data Management Learning

Based on *DAMA-DMBOK2*, Chapter 4

This deck is based on the uploaded *DAMA-DMBOK: Data Management Body of Knowledge, 2nd Edition*, especially Chapter 4 on Data Architecture.

- Understand what DAMA means by **data architecture** and why it matters.
- See how data architecture links **business strategy**, **data requirements**, and **technology execution**.
- Recognize core artifacts: **enterprise data models**, **subject area models**, and **data flows**.
- Identify practical activities, governance needs, and success factors for implementation.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

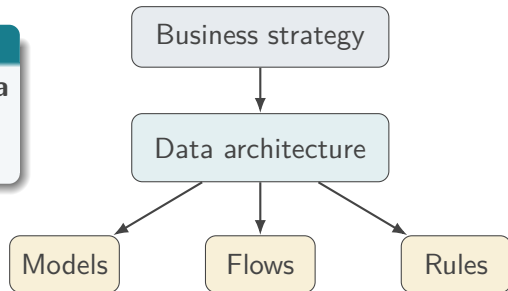
What is data architecture?

Core idea

Data architecture is the **blueprint for managing data assets**. It describes how data is organized, defined, moved, controlled, and aligned with enterprise goals.

- Covers artifacts, activities, and behaviors
- Works across multiple levels of abstraction
- Helps people understand data that is too large and complex to grasp directly

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).



Why organizations need it

- Bridges **business strategy** and **technology execution**
- Helps organizations evolve products, services, and data faster
- Makes sure processes consistently get the data they need
- Improves alignment between **business** and **IT**
- Supports agility, integration, and controlled change

Realistic examples

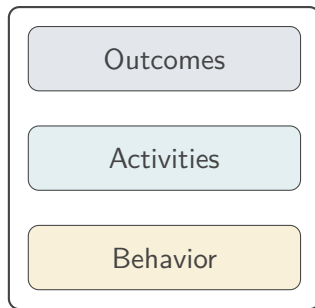
- Bank launching a new digital lending journey
- Hospital integrating patient, lab, and billing data
- Manufacturer shifting from selling products to *pay-per-use* services

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

DAMA's 3-part view of data architecture

Three components

- 1 **Outcomes** – models, definitions, data flows, roadmaps
- 2 **Activities** – forming, deploying, and maintaining architecture
- 3 **Behavior** – collaboration, mindset, and architectural discipline



Together form data architecture

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Key outcomes and practice goals

- Define the **current state** of organizational data
- Provide a **standard business vocabulary** for data and components
- Align data architecture with **enterprise strategy** and **business architecture**
- Express **strategic data requirements** and high-level integrated designs
- Connect architecture work to the overall enterprise roadmap

What good practice looks like

Use architecture artifacts as master blueprints, influence projects and stakeholders, and establish the semantics of the enterprise through shared vocabulary.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Architecture works across domains

Domain	Main focus	Example question
Business architecture	How the enterprise creates value	Which capabilities and processes matter most?
Data architecture	How data should be organized and managed	What data do we need, and how should it fit together?
Applications architecture	Structure and function of applications	Which systems create or consume the data?
Technology architecture	Platforms, networks, and infrastructure	What technical environment hosts and protects the systems?

Takeaway

Data architecture does not stand alone. It must work with business, application, and technology architecture.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Enterprise architecture frameworks

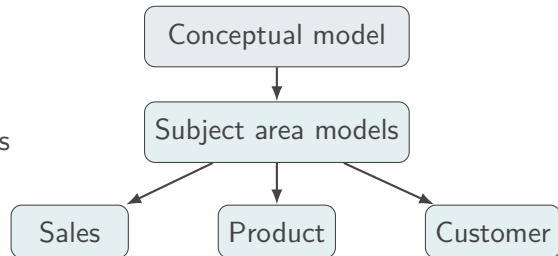
- Frameworks help non-architects understand architecture relationships
- DAMA references common enterprise architecture frameworks
- The most visible example in the chapter is the **Zachman Framework**
- Zachman shows what models should exist from multiple stakeholder perspectives

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

	What	How	Where	Who	When	Why
Build						
Design						
Own						
Planner						

Enterprise data model: the master blueprint

- The most detailed architecture design document is the **enterprise data model**
- It contains names, definitions, conceptual and logical entities, relationships, and business rules
- It is built across levels: conceptual → subject area → logical detail
- Physical models belong more directly to data modeling and design



Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Subject area models make the enterprise understandable

- Each entity should live in one subject area, but can relate to entities in other areas
- Subject areas can be formed using normalization rules, ownership structures, value chains, or business capabilities
- DAMA notes that a **combined top-down and bottom-up approach** is often practical

Example subject areas

Customer, Product, Sales Order, Inventory, Supplier, Billing, Service, Claims.

Teaching example

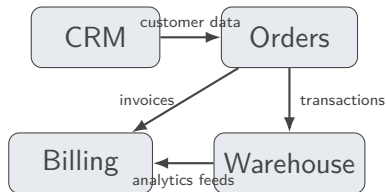
In retail, *Customer* belongs in one core subject area, but connects to orders, returns, support cases, and loyalty events across the enterprise.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Data flow design and lineage

- Data flows show how data moves through processes and systems
- They reveal origin, storage, use, and transformation
- They can map relations between applications, data stores, networks, roles, and locations
- They support lineage analysis and better control of change

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).



Two useful ways to show data flow

1. Matrix view

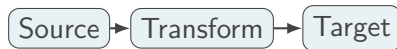
Best for showing which processes create, read, update, or depend on major entities.

	Customer	Order
Sales	C/R	C/R
Billing	R	R/U
Support	R/U	R

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

2. Diagram view

Best for showing system-to-system movement and transformation of data.



Mini discussion

A university wants one trusted student record across admissions, exams, finance, and alumni systems.

- ❶ What should be the core subject areas?
- ❷ Where would the main data ownership conflicts appear?
- ❸ Which data flows would you document first?

Possible starting answer: Student, Course, Enrollment, Payment, Degree, Identity, and Communication.

Activities: establish a data architecture practice

- Evaluate existing architecture specifications
- Develop a roadmap for target-state architecture
- Manage enterprise data requirements within projects
- Use architecture to reduce complexity and avoid uncontrolled system growth

Why this matters

Without active architecture practice, systems drift into copy-heavy, interface-heavy, low-trust environments — what teams often call *data spaghetti*.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

What architects actually do in practice

Activity		Practical meaning
Evaluate current state		Review models, interfaces, definitions, redundancies, and gaps
Roadmap target state		Decide the future blueprint, priorities, and transition path
Guide projects		Make sure new initiatives follow enterprise data requirements
Coordinate across domains		Align with business, application, and technology architects

Example

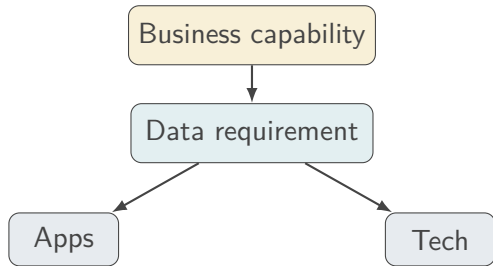
When a telecom launches a self-care app, architects decide which customer, product, and billing data must stay authoritative and how the app should consume them.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Integrating data architecture with enterprise architecture

- Data architecture must be integrated, not isolated
- Business capabilities drive data dependencies
- Application and technology choices must respect data requirements
- Architecture decisions should be visible in enterprise governance forums

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).



Tools and techniques from the chapter

Tools

- Data modeling tools
- Asset management software
- Graphical design applications

Techniques

- Lifecycle projections
- Diagramming clarity

Simple message

The tool does not create architecture quality by itself. Clear diagrams, consistent semantics, and disciplined lifecycle thinking do.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Implementation guidelines and common risks

- DAMA explicitly highlights **readiness assessment** and **risk assessment**
- Implementation depends on organization and culture, not only design quality
- Architecture work needs communication, adoption, and behavior change
- Success requires balancing enterprise standards with local project realities

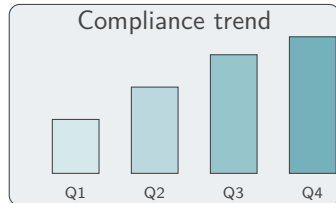
Typical risks

Poor buy-in, unclear ownership, weak standards adoption, over-complex diagrams, and projects bypassing enterprise architecture.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

- Data architecture needs governance to stay relevant and enforced
- Standards compliance is a major governance concern
- Metrics in the chapter include:
 - architecture standards compliance rates
 - trends in implementation
 - business value metrics

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).



Real-world style examples by section

Section		Example
Business drivers		Automotive firm adds connected-car services and now needs operational vehicle data as a managed asset
Enterprise data model		Insurer defines Party, Policy, Claim, Payment, and Product as shared enterprise concepts
Subject areas		University separates Student, Course, Enrollment, Finance, and Identity but keeps them related
Data flows		Hospital maps lab results from source system to EHR, billing, and analytics environments
Governance metrics	&	Retail group tracks compliance of new project schemas with enterprise customer definitions

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Scenario

A company wants speed. Project teams want to copy customer data into every application because it is easier in the short term.

- What short-term benefit are they chasing?
- What long-term data architecture risks appear?
- Where would you allow duplication, and where would you not?

Hint: Use the language of trust, redundancy, lineage, integration cost, and ownership.

Key takeaways

- ➊ Data architecture is the enterprise **blueprint** for managing data assets.
- ➋ It connects **business strategy** to **data requirements** and then to **technology execution**.
- ➌ Its main artifacts are enterprise data models, subject area models, and data flow designs.
- ➍ It works only when integrated with broader enterprise architecture and governance.
- ➎ Good architecture reduces complexity, improves trust in data, and increases business agility.

Source: DAMA-DMBOK2, Chapter 4 (Data Architecture).

Questions, reflections, or examples from your own organization?

Possible prompts:

- Where does your organization struggle most: models, flows, ownership, or standards?
- Which enterprise subject areas are hardest to define consistently?
- What architecture artifact would create the fastest business value for you?

Primary source used for the content structure and DAMA-aligned concepts in this deck:

DAMA International. *DAMA-DMBOK: Data Management Body of Knowledge, Second Edition*. Technics Publications, 2017. Chapter 4: Data Architecture.

Slides in this deck specifically reflect the chapter's definition of Data Architecture, its business drivers, architecture domains, enterprise data models, subject area models, data flows, activities, tools, techniques, implementation guidance, governance, and metrics.